Adani University

Faculty of Engineering Sciences and Technology (FEST)

Department of Computer Science and Engineering (AI-ML)

PROJECT-BASED LEARNING (PBL) & RESEARCH IMPLEMENTATION

Course: ECSCT24301 – Foundation of AI

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1. Project Title

End-to-End Artificial Intelligence Problem-Solving: From Problem Formulation to Intelligent Agent Implementation

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2. Project Objective

The primary objective of this project is to provide students with hands-on experience in formulating, designing, implementing, and evaluating classical and modern Artificial Intelligence (AI) solutions for non-trivial real-world problems.

Rather than treating AI algorithms as black boxes, students are required to rigorously model the problem environment, define state spaces or knowledge bases, select and implement appropriate heuristic search, adversarial reasoning, constraint satisfaction, probabilistic inference, or automated planning algorithms, and evaluate computational efficiency and optimality.

Students may choose between two tracks:

1. **Applied AI Project Track:** Formulate an engineering/real-world problem, model the agent environment, implement algorithms or knowledge-based systems, and deploy or demonstrate a functioning prototype.
2. **Research Implementation Track:** Select, replicate, and critically analyze a peer-reviewed research paper from the last five years (2020–2025) covering fundamental AI search, logic, planning, or reasoning techniques, and benchmark against baseline methods.

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3. Project Team and Constraints

- **Team Size:** The project must be completed in teams of **2 to 3 students**. Individual submissions or teams exceeding 3 members are strictly disallowed.

- **Batch Constraint:** Inter-batch teams are **not permitted**. All team members must belong to the same academic division/batch.
- **Topic Uniqueness:** Duplicate problems or identical research papers across different teams within the same batch will not be approved. Topics are allocated on a strictly first-come, first-reviewed basis.
- **Citation Requirement:** Every external resource, library, baseline code, dataset, and reference paper must be fully cited following APA style guidelines.

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4. Track Selection & Problem Scoping

Track A: Applied AI System Development

Teams model a practical real-world task as an intelligent agent problem. Projects must feature:

- Formal environment and PEAS (Performance Measure, Environment, Actuators, Sensors) specification.
- Problem formulation: state representation, initial state, successor function, goal test, and path cost function .
- Implementation of suitable core AI algorithms (e.g., A^*\ search with admissible and consistent heuristics, Minimax with depth-limited evaluation functions, CSP solvers with arc consistency, or Bayesian inference).

Track B: AI Research Paper Implementation

Teams reproduce an algorithmic advancement or benchmarking study published in recent literature. Projects must feature:

- A comprehensive literature review synthesizing at least 3–5 related papers.
- Re-implementation of the proposed algorithm from scratch or extending official open-source codebases with novel experiments.
- Comparative experimental evaluation against established classical baselines (e.g., comparing a novel heuristic against standard Manhattan distance in).

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5. Suggested Project Domains & Themes

Domain	Indicative Project Topics / Challenges	Relevant Syllabus Unit

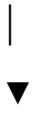
Autonomous Navigation & Routing	Dynamic path planning with moving obstacles (, -Lite, Bidirectional Search); Multi-agent path finding (MAPF).	Unit 1 & 2
Game Playing & Adversarial Search	Intelligent agents for Connect Four, Chess variants, Othello, or Gomoku using Minimax, Alpha-Beta pruning, and Monte Carlo Tree Search (MCTS).	Unit 2
Optimization & Scheduling (CSP)	University timetable scheduling, crew roster scheduling, Sudoku/N-Queens variant solvers using Backtracking with MRV, Degree Heuristic, and AC-3.	Unit 2
Robotic Manipulation & Planning	Automated warehouse package sorting or classic Blocks World solving using Goal-Stack Planning and Partial-Order Planning (POP).	Unit 4
Diagnostic & Expert Systems	Rule-based clinical diagnostic support (inspired by MYCIN), fault diagnosis in electrical grids using certainty factors or forward/backward chaining.	Unit 3 & 4
Uncertainty & Decision Support	Probabilistic medical triage or fraud risk estimation using Bayesian Networks and Belief Updating; Fuzzy Logic control systems for HVAC or energy grids.	Unit 3
Knowledge Representation	Semantic Web ontologies, frame-based systems, and knowledge graphs for automated question-answering over structured technical manuals.	Unit 3

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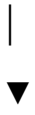
6. The End-to-End AI Engineering Pipeline

Every team must execute and document the following development stages:

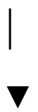
[Problem Definition & PEAS Formulation]



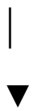
[State-Space / Knowledge-Base Representation]



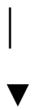
[Algorithm Selection & Heuristic/Rule Design]



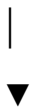
[Baseline Implementation & Optimization]



[Empirical Benchmarking & Complexity Analysis]



[Error & Failure Mode Analysis]



[Interactive Demonstration & Deployment]

Required Pipeline Stages:

1. **Problem Definition:** Clear objective, scope, and PEAS characterization.
2. **Environment & Agent Modeling:** Precise mathematical formulation (states, actions, transition model, goal condition, cost metrics).
3. **Knowledge Base / Data Acquisition:** Collection and validation of test instances, benchmark maps, or datasets (Kaggle/UCI).
4. **Baseline Model:** Implementation of a fundamental search (BFS/DFS) or naive random/greedy approach to establish a performance benchmark.
5. **Advanced Model Development:** Implementation of informed, adversarial, CSP, probabilistic, or planning algorithms.
6. **Heuristic & Optimization Engineering:** Formulation of admissible/consistent heuristics (), constraint propagation strategies, or domain rules.
7. **Empirical Benchmarking:** Measuring execution time, nodes generated/expanded, memory footprint, path optimality, and win rates across varying problem scales.
8. **Error & Failure Analysis:** Investigation of edge cases where the algorithm fails, times out, or produces suboptimal paths; discussion of state-space explosions.

9. **Interactive Prototype / Demonstration:** A command-line visualization, GUI (Pygame/Streamlit/Tkinter), or interactive notebook demonstrating execution on unseen inputs.
10. **Conclusion & Future Directions:** Critical reflection on algorithmic limitations, space complexity bottlenecks, and scalability.

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7. Project Milestones and Course Schedule Alignment

Phase	Course Timeline	Expected Deliverables	Weightage
Stage 1: Proposal & Topic Approval	**3rd Week of August**	Team formulation, problem definition, track selection (Project vs. Research), PEAS description, dataset/literature source, and proposed methodology.	**10%**
Stage 2: Design & Initial Results	**4th Week of September**	State-space representation, baseline implementation (uninformed search / basic rules), initial experimental results, and heuristic design.	**15%**
Mid-Semester Review	**1–31 October**	Progress presentation, core algorithm implementation (e.g., Minimax, CSP, or Bayesian model), code walk-through, and intermediate report.	**20%**
Stage 3: Advanced Experiments & Tuning	**2nd Week of November**	Comparative benchmarking, heuristic verification, sensitivity/scalability tests, GUI/demo integration, and draft report.	**25%**
Stage 4: Final Submission & Viva Voce	**3rd Week of November**	Final report (Word/PDF), GitHub repository submission, live software demonstration, and individual oral defense (Viva Voce).	**30%**

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8. Project Proposal Format (Stage 1 – Due 3rd Week of August)

The proposal must be submitted as a 2–3 page document containing:

1. **Header Details:** Course Code, Academic Year, Team Members (Names, Enrollment Numbers, Email IDs).
2. **Project Title & Track:** Applied AI Project Track OR Research Implementation Track.
3. **Problem Statement & Motivation:** Detailed real-world context and operational significance.
4. **Agent & Environment Characterization (PEAS):** Table specifying Performance Measure, Environment, Actuators, and Sensors.
5. **Formulation Details:** Initial State, Action Set, Goal State, and Cost Metric.
6. **Data / Literature Sourcing:** Kaggle/UCI dataset URL, benchmark generator, or reference research paper citation (APA format).
7. **Proposed Algorithmic Approaches:** Proposed baseline and advanced algorithms/heuristics to be explored.
8. **Planned Tooling & Libraries:** Programming language, simulation frameworks, and GUI tools.

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9. Final Project Report Structure

The final project report must be neatly formatted, written in clear academic prose, and structured as follows:

- **Title Page:** University name, department, course code, project title, supervisor name, team members with enrollment numbers, and submission date.
- **Abstract / Executive Summary:** Concise overview (250–300 words) summarizing the problem, implemented algorithms, key experimental results, and main takeaways.
- **1. Introduction & Background:** Real-world significance, scope, and foundational AI principles utilized.
- **2. Problem Formulation & Environment Modeling:** PEAS description, environment properties, formal mathematical definition of the state space, transition model, and goal test.
- **3. Literature Review / Related Work:** (Mandatory for Research Track; minimum 5 recent papers covering prior solutions).
- **4. Methodology & Algorithmic Architecture:**
    - Theoretical explanation of chosen techniques.

    - Heuristic function mathematical formulations and consistency proofs.
    - Pseudocode and architectural flowcharts.
- **5. Implementation Details:** Tech stack, key data structures, state indexing, and edge case handling.
- **6. Experimental Design & Setup:** Benchmark problem sets, map topologies, branching factors, and test harness details.
- **7. Results & Comparative Analysis:** Quantitative benchmark tables, convergence plots, node expansion charts, and runtime comparisons.
- **8. Error Analysis & Scalability Bottlenecks:** Analysis of suboptimal decisions, worst-case memory blowups, and time-out conditions.
- **9. Interactive System Demonstration:** Screen captures and functional walk-through of the demo interface.
- **10. Conclusion & Future Work:** Summary of findings, real-world deployment challenges, and potential extensions.
- **References:** Sourced following the **American Psychological Association (APA)** referencing standard.

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10. Results to be Reported in the Final Deliverable

- Comprehensive **Benchmark Summary Table** comparing all evaluated algorithms/heuristics across small, medium, and large instances.
- Comparative graphical plots:
    - State space size / Depth vs. Execution Time ().
    - Problem instance complexity vs. Number of Generated/Expanded Nodes.
    - Search tree pruning efficiency (for game playing / adversarial projects).
- Formal statement of heuristic properties () with counterexamples explored during trial phases.
- Demonstration logs illustrating exact step-by-step decision sequences for sample runs.

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11. Assessment Scheme

In alignment with the Continuous Evaluation policy, projects will be graded on the following rubric:

Assessment Component	Criteria	Weightage
1. Problem Formulation & Literature Review	Precision of problem scope, mathematical modeling of state space/logic, PEAS formulation, and depth of literature review.	**10%**
2. Algorithmic Design & Implementation	Correctness, elegance, and modularity of implemented AI algorithms, heuristic functions, or reasoning engines.	**20%**
3. Baseline & Comparative Evaluation	Implementation of proper baselines, comprehensive experimental benchmarking, and accuracy of comparison tables.	**20%**
4. Heuristic Analysis & Optimization	In-depth exploration of heuristic admissibility, pruning techniques, and computational complexity tradeoffs.	**15%**
5. Technical Report Quality	Completeness, academic writing quality, proper structure, and strict adherence to APA citations.	**15%**
6. Software Demonstration & UI	Working prototype, responsiveness, error-handling, and interactive visualization of search/reasoning steps.	**10%**
7. Individual Viva Voce & Defense	Subject knowledge, ability to defend design decisions, understanding of algorithmic nuances, and individual contributions.	**10%**
Total		**100%**

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12. Viva Voce – Areas of Individual Assessment

During the final oral defense, each student will be examined individually on:

- Detailed understanding of state space representations and the branching factor of the problem.
- Theoretical properties of the chosen algorithms (Completeness, Optimality, Time Complexity , Space Complexity).

- Mathematical proof or empirical demonstration of heuristic admissibility () and consistency ().
- Practical rationale behind selecting specific evaluation metrics and test benchmarks.
- Explanation of algorithmic behavior in edge cases, dead ends, or cycles.
- Direct walkthrough of self-authored code sections.

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13. Final Presentation and Demonstration Structure

The final presentation will consist of a **15-minute presentation + 5-minute live demo + 5-minute Q&A**:

1. **Slide 1–2:** Problem Statement, Real-world Motivation, and PEAS Model.
2. **Slide 3–4:** State Space, Action Operators, and Knowledge Representation.
3. **Slide 5–6:** Core Algorithms, Heuristic Design, and Theoretical Complexity.
4. **Slide 7–8:** Experimental Setup and Benchmark Test Instances.
5. **Slide 9–10:** Comparative Performance Results (Charts & Comparison Table).
6. **Slide 11:** Error Analysis, Failure Modes, and Scalability Limits.
7. **Slide 12:** Conclusion, Ethical Implications, and Future Scope.
8. **Live Software Demonstration:** Real-time run of the agent/solver on randomly generated or user-defined inputs.
